

Introduction to Number Theory

Exercise 1. Answer: E. Statement “any prime number p equals to the product of its divisors” is true since the only divisors of p are 1 and p by definition.

Exercise 2. Answer: D. Statement $\text{lcm}(n+1, n-1) = n^2 - 1$ is false. For instance, take $n := 5$. Then, $\text{lcm}(6, 4) = 12 \neq 5^2 - 1 = 24$.

Exercise 3. Answer: C. Answer is 9. Indeed, $9 \cdot 5 = 45 \equiv 1 \pmod{11}$.

Exercise 4. Answer: C. Answer is 15. Indeed, $\text{gcd}(15, 21) = 3 \neq 1$, so $15 \notin \mathbb{Z}_{21}^\times$ by definition.

Exercise 5. Answer: A. Notice that $n^2 + 1 = n(n+1) - (n-1)$. Thus, if $(n+1) \mid (n^2 + 1)$ we should have $(n+1) \mid (n-1)$ which is possible only if $n-1 = 0$. Thus, $n = 1$ is the only such number.

Exercise 6. Answer: C. Let us find roots of f . Notice that:

$$f(x) = 2x^2 - (3p + q)x + (p^2 + pq)$$

By equating $f(x) = 0$ we get two solutions: $r_1 = p$ and $r_2 = \frac{p+q}{2}$. One can easily check that the only possible p and q for r_2 to be prime is $(7, 3)$. Thus, the sum is 10.

Exercise 7. Answer: B. We have the only $\alpha_1 = 100$ which by the definition of τ function gives $\tau(p^{\alpha_1}) = \alpha_1 + 1 = 101$.

Exercise 8. Answer: D. Suppose $n = \prod_{j=1}^r p_j^{\alpha_j}$. Then, $\tau(n) = \prod_{j=1}^r (\alpha_j + 1) = 7$. The only possible way to represent 7 as a product of integers is to have the only non-zero α such that $\alpha + 1 = 7$. Thus, $\alpha = 6$, so $n = p^6$ for arbitrary prime p . Therefore, $\tau(n^2) = \tau(p^{12}) = 13$, following the reasoning from the previous exercise.

Exercise 9. Answer: B. Suppose $m = \prod_{j=1}^r p_j^{\alpha_j}$, so again $\tau(m) = \prod_{j=1}^r (\alpha_j + 1)$. For this expression to be odd, we need every term $\alpha_j + 1$ to be odd. This is possible only if every α_j is even. Therefore, m must be a square of a number. The smallest square greater than n^2 is of course $(n+1)^2$.

Introduction to Abstract Algebra

Exercise 1. Answer: E. Notice that if $xy = 1$ then $y = \frac{1}{x}$. If x is an integer then y is integer only if $x = \pm 1$. Therefore,

$$X \cap \mathbb{Z}^2 = \{(1, 1), (-1, -1)\}, \quad X \cap \mathbb{N}^2 = \{(1, 1)\}.$$

Therefore, first two statements are correct. When it comes to the third statement, it is also true: indeed, the identity element is $(1, 1)$ while the inverse element for (x_1, x_2) is $(1/x_1, 1/x_2)$ (such operation is closed since $(1/x_1)(1/x_2) = 1/(x_1 x_2) = 1$). Other properties are also satisfied.

Exercise 2. Answer: A. Identity element in this group is 1. Indeed, $a \oplus 1 = 1 \oplus a = a + 1 - 1 = a$. Associativity follows immediately from the sum operation associativity. Inverse element for a is $2 - a$ since:

$$a \oplus (2 - a) = a + 2 - a - 1 = 1$$

All operations are closed since the sum and subtraction of two integers is an integer.

Exercise 3. Answer: C. Suppose the identity element $E \in \mathbb{R}^2$ exists. Then, $A \oplus E = E$ by definition. However, the midpoint of AE is E only if $A = E$. Since identity element depends on the chosen group element A , it does not exist.

Exercise 4. Answer: E. Indeed, take $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. Suppose inverse B exists. Then, $AB = E_{2 \times 2}$ where $E_{2 \times 2}$ is an identity matrix. Therefore, $\det(AB) = \det A \cdot \det B$. Since $\det A = 4$ we have $\det B = \frac{1}{4}$. However, the determinant of a matrix, composed of integers, cannot be a fraction. Therefore, the inverse does not exist for A .

Exercise 5. Answer: A. Let us check all the properties carefully:

- **Closure:** Take $f, g \in \mathcal{F}$. The function $f \times g$ has a domain Ω and range $\mathbb{G}$ since $f(\omega) \times g(\omega)$ always outputs $\mathbb{G}$ for any $\omega \in \Omega$ due to the closure of $\mathbb{G}$.
- **Identity Element:** Define identity element $i \in \mathcal{F}$ as a function $i(\omega) = e$ for any $\omega \in \Omega$ where e is an identity element in $\mathbb{G}$. Then:

$$(f \star i)(\omega) = f(\omega) \times i(\omega) = f(\omega) \times e = f(\omega) \implies f \star i = f$$

- **Inverse:** Define inverse element for $f \in \mathcal{F}$ as g defined as $g := (f(\omega))^{-1}$ for every $\omega \in \Omega$. Then:

$$(f \star g)(\omega) = f(\omega) \times (f(\omega))^{-1} = e \quad \forall \omega \in \Omega \implies f \star g = i$$

- **Associativity:** Holds due to the associativity of $\times$.
- **Commutativity:** Holds due to the commutativity of $\times$.

Exercise 6. Answer: A. First, $f : x \mapsto x^2$ is obviously closed over $[0, 1]$. Let us check why it is an automorphism. First, check whether it is a homomorphism:

$$f(x \cdot y) = (x \cdot y)^2 = x^2 \cdot y^2 = f(x) \cdot f(y)$$

Now, let us show that f is a bijection. Indeed, f is injective since $f(x) = f(y)$ implies $x^2 = y^2$ which is possible only if $x = y$ when both x and y are positive. f is surjective since for any $y \in [0, 1]$ there is $x = \sqrt{y}$ such that $f(x) = y$. Therefore, f is an automorphism.

Exercise 7. Answer: C. In fact, $\varphi_1 = \det$ while $\varphi_2 = \text{trace}$. It is well-known that $\det(AB) = \det A \det B$ and $\text{trace}(A+B) = \text{trace}(A) + \text{trace}(B)$. Therefore, both functions are homomorphisms.

Exercise 8. Answer: A. Let us check the homomorphism property. Take $a + bi, c + di \in \mathbb{C} \setminus \{0\}$. Then:

$$\psi((a + bi)(c + di)) = \psi((ac - bd) + (ad + bc)i) = \begin{bmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{bmatrix}$$

On the other hand:

$$\psi(a + bi)\psi(c + di) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{bmatrix}$$

Therefore, $\psi((a+bi)(c+di)) = \psi(a+bi)\psi(c+di)$, so ψ is a homomorphism. Let us show it is a bijection. Suppose $\psi(a+bi) = \psi(c+di)$. Then:

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} c & -d \\ d & c \end{bmatrix} \implies a = c, b = d$$

Thus, ψ is injective. To show the surjectivity notice that for any matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ there is $a+bi$ such that $\psi(a+bi) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$. Therefore, ψ is a bijection. Therefore, ψ is an isomorphism.

Exercise 9. Let $\mathbb{G}$ be a group of order 9. If $\mathbb{G}$ contains the element of order 9, then it is cyclic and thus abelian, so we are done. Suppose this is not the case and we have the element $x \in \mathbb{G}$ of order other than 9. Due to Lagrange's theorem, the order of g must divide the order of $\mathbb{G}$, so it is either 1, 3 or 9. Since 1 corresponds to the identity element, suppose $\mathbb{H}_X = \langle x \rangle$ has order 3. Let $y \in \mathbb{G} \setminus \mathbb{H}_X$ be some other non-identity element of $\mathbb{G}$ and $\mathbb{H}_Y := \langle y \rangle$. Since $\mathbb{H}_X \cap \mathbb{H}_Y$ is a subgroup of $\mathbb{H}_X$ and $|\mathbb{H}_X| = 3$ by Lagrange's Theorem either $\mathbb{H}_X \cap \mathbb{H}_Y = \{e\}$ or equals $\mathbb{H}_X$. Since $y \notin \mathbb{H}_X$, we conclude that $\mathbb{H}_X \cap \mathbb{H}_Y = \{e\}$. Therefore,

$$|\mathbb{H}_X \mathbb{H}_Y| = \frac{|\mathbb{H}_X| \cdot |\mathbb{H}_Y|}{|\mathbb{H}_X \cap \mathbb{H}_Y|} = 9$$

Therefore, $\mathbb{G} = \mathbb{H}_X \mathbb{H}_Y = \{e, x, x^2, y, y^2, xy, xy^2, x^2y, x^2y^2\}$ since $x^3 = y^3 = e$. Now, consider $yx \in \mathbb{G}$. It must be one of elements listed in $\mathbb{G}$:

- $yx \neq e$, otherwise $y = x^2$ but $\mathbb{H}_X \cap \mathbb{H}_Y = \{e\}$.
- $yx \neq x$, otherwise $y = e$.
- $yx \neq x^2$, otherwise $x = y$.
- $yx \neq y$, otherwise $x = e$.
- $yx \neq y^2$, otherwise $x = y$.
- $yx \neq xy^2$, otherwise $xyx = x$ (multiplied both sides by y), but $(yx)^3 = yx(yxy)x = yx^3 = y = e$, so $y = e$.
- $yx \neq x^2y$, otherwise $xyx = y$ and thus $x = e$ (see explanation above).
- $yx \neq x^2y^2$, otherwise $(yx)^2 = (yx)(yx) = (x^2y^2)(yx) = e$. Yet, $(yx)^3 = e$, so $yx = e$ or $y = x^2$, contradicting $\mathbb{H}_X \cap \mathbb{H}_Y = \{e\}$.

Therefore, $yx = xy$, so $\mathbb{G}$ is an abelian group.

Polynomials

Exercise 1. Answer: A. Note that $\deg(p - q) = 4$ since p and q are of different degrees. Therefore, $\deg((p - q)r) = \deg(p - q) + \deg(r) = 4 + 5 = 9$.

Exercise 2. Answer: D. It suffices to plug each of $\{0, 1, 2, 3, 4\}$ in $4x^2 + 7$ and verify that the result is never 0 over $\mathbb{F}_5$.

Exercise 3. Answer: A. Note that $p(x) = a(x - 1)(x - 2)$. Since $p(0) = 2$, we have $a = 1$. Therefore, $p(x) = x^2 - 3x + 2$ and thus $b = -3$, $c = 2$. The sum $a + b + c$ is then 0.

Exercise 4. Answer: C. Roots of Q are $\pm i$. If $Q(x) \mid P(x)$, we must have $P(\pm i) = 0$. Thus we have $P(i) = i^{2n} + i^n - 2 = (-1)^n + i^n - 2$. Consider four cases:

- $n = 4k$. Then, $P(i) = (-1)^{4k} + i^{4k} - 2 = 0$.
- $n = 4k + 1$. Then, $P(i) = (-1)^{4k+1} + i^{4k+1} - 2 = -3 + i \neq 0$.
- $n = 4k + 2$. Then, $P(i) = (-1)^{4k+2} + i^{4k+2} - 2 = -2 \neq 0$.
- $n = 4k + 3$. Then, $P(i) = (-1)^{4k+3} + i^{4k+3} - 2 = -3 - i \neq 0$.

Therefore, n must be divisible by 4. It is easy to check that in such case $P(-i)$ equals 0 as well.

Exercise 5. Answer: A. Since polynomial has d distinct roots, the probability of choosing a root is *exactly* d/p . The probability that Alice fools Bob after n rounds is then $(d/p)^n$.

Exercise 6. Answer: B. According to the Schwartz-Zippel Lemma, we have $\Pr[f(\mathbf{x}) = 0 \mid \mathbf{x} \xleftarrow{R} \mathbb{S}] \leq \deg f / |\mathbb{S}|$, so we need to find the size $|\mathbb{S}|$. Our claim is that the number of solutions to $x_1 + \dots + x_n = 1$ over $\mathbb{F}_p$ is exactly p^{n-1} . Indeed, for any p^{n-1} possible values of $(x_2, \dots, x_n)$ there is a unique x_1 such that $x_1 + \dots + x_n = 1$, which is $1 - \sum_{i=2}^n x_i$. Therefore, $\Pr[f(\mathbf{x}) = 0] \leq \deg f / p^{n-1}$.

Field Extensions

Exercise 1. Answer: D. Since $i^2 = -1$, we have $(3+i)(4+i) = 12 + 7i + i^2 = 11 + 7i$. Reducing this modulo 7 gives 4.

Exercise 2. Answer: D. Note that $2/i = 2i/i^2 = 2i/(-2) = -i = (p-1)i$.

Exercise 3. Answer: A. Such field extension is valid as long as $v^2 + v + 1$ has no roots over $\mathbb{F}_p$. Note that $v^2 + v + 1 = 0$ has no solutions if and only if $v^3 - 1$ has no trivial solutions: indeed, $v^3 - 1 = (v-1)(v^2 + v + 1)$. From the problem statement we know this is the case when $p \equiv 1 \pmod{3}$. From the provided choice, this is the case for $p = 8431$.

Exercise 4. Answer: B. Field extension $\mathbb{F}_5[i]/(r(i))$ is valid only when $r(x)$ is irreducible over $\mathbb{F}_5$. This is the case when $r(x) = x^2 + 2$.

Exercise 5. Answer: A. The zero of the provided polynomial is:

$$x_0 = \frac{i+3}{i} = \frac{i^2+3i}{i^2} = 1 + \frac{3i}{-2} = 1 - 3i \cdot 3 = 1 - 9i = 1 + i$$

Exercise 6. Answer: A. Notice that $j^4 = (j^2)^2 = \xi^2$. Since $\xi = 1 + i$ by definition, we have $j^4 = (1 + i)^2 = -1 + 2i = 4 + 2i$.

Exercise 7. Answer: C. Again, using identity $j^2 = \xi$, we arrive at $j^3 + 2i^2\xi = j\xi - 4\xi = j\xi + \xi$. Therefore, both the real and imaginary parts are same and equal to ξ .

Exercise 8. Answer: B. Let us expand the expression: $(a_0 + a_1j)b_1j = a_0b_1j + a_1b_1j^2 = a_1b_1\xi + a_0b_1j$. So the real part is thus $a_1b_1\xi$.

Exercise 9. Answer: B. $z\bar{z} = (c_0 + c_1j)(c_0 - c_1j) = c_0^2 - c_1^2j^2 = c_0^2 - \xi c_1^2$.

Elliptic Curves and Pairing

Exercise 1. Answer: 1. If $(2, 3) \in E/\mathbb{F}_7$, then $3^2 = 2^3 + b$ and thus $b = 9 - 8 = 1$.

Exercise 2. Answer: A. If $P \oplus Q = \mathcal{O}$, then P and Q must have the same x coordinates and opposite y coordinates. Points $P = (2, 3)$ and $Q = (2, 8)$ satisfy this condition.

Exercise 3. Answer: A. According to Hasse's theorem, we have

$$r = p^2 + 1 - t, \quad |t| \leq 2p$$

In our case, $p = 167$, so $r = 167^2 + 1 - t$ for $|t| \leq 334$. The only suitable option among provided is $167^2 - 5$.

Exercise 4. Answer: C. Upper bound for factors of $|E| = qr$ is $\max\{q, r\}$, so the complexity is $\mathcal{O}(\sqrt{\max\{q, r\}})$.

Exercise 5. Answer: A. Note that reflexivity is not satisfied. Indeed, $a + a < 0$ does not hold for $a \in \mathbb{Q}_{>0}$.

Exercise 6. Answer: C. By definition, $[1.4]_{\sim} = \{x \in \mathbb{R} : x - 1.4 \in \mathbb{Z}\}$. That means that every $x \in [1.4]_{\sim}$ is of form $x = 1.4 + n$ where $n \in \mathbb{Z}$. This can be further simplified to $x = 0.4 + n'$ with $n' \in \mathbb{Z}$. Thus, the fractional part must be 0.4.

Exercise 7. Answer: D. For homogeneous projective coordinates we must have $X_2/X_1 = Y_2/Y_1 = Z_2/Z_1$. In our case, the fraction $X_2/X_1 = 16/4 = 4$ while $Y_2/Y_1 = 8/3$, so two points are not equivalent.

Exercise 8. Answer: E. Projective coordinates are effective since addition of points over the projective space does not require inversion. The only inversion needed is during the transforming point back to the affine space.

Exercise 9. Answer: D. The number of additions equals the number of non-zero bits in the binary representation of k . Since $19 = 10011_2$, the number of additions is 3.

Exercise 10. Answer: A. One way to reduce the number of inversions is to calculate the $t \leftarrow (a + b)^{-1}$ first, which gives $(a + b)^{-4} = t^4$ for free. All left to do is to calculate $(a^2 + c^2)^{-1}$, making the total cost 2 inversions. However, we can use only one inversion: notice that

$$\frac{a - b}{(a + b)^4} + \frac{c}{a + b} + \frac{d}{a^2 + c^2} = \frac{(a - b)(a^2 + c^2) + c(a + b)^3(a^2 + c^2) + d(a + b)^4}{(a + b)^4(a^2 + c^2)}$$

Therefore, we need only one inversion: first we calculate $z \leftarrow (a + b)^4(a^2 + c^2)$ and then calculate z^{-1} . Then we multiply by the numerator, costing zero inversions. So the optimal solution is 1 inversion. Note that the second approach requires larger number of field multiplications, so it might be not optimal in case inversion is cheap.

Exercise 11. Answer: B. Notice that $e([4]G_1, [4]G_2) = e(G_1, G_2)^{4 \cdot 4} = e(G_1, G_2)^{16} \neq e([3]G_1, [5]G_2) = e(G_1, G_2)^{15}$.

Exercise 12. Answer: E. Note that $e(P, P)e(Q, Q) = e(G, G)^{x^2}e(G, G)^{y^2} = e(G, G)^{x^2+y^2}$. Therefore, checking whether $e(P, P)e(Q, Q) = e(G, G)$ is equivalent to checking whether $x^2 + y^2 = 1$.

Commitment Schemes

Exercise 1. Answer: C. We have $H(x) = C$. By the definition of H , we have $13x + 17 = 39$ over $\mathbb{F}_{41}$. Therefore, $13x = 22$, so $x = 22 \cdot 13^{-1} = 8$.

Exercise 2. Answer: A. If $U = [6]G$ then the Pedersen commitment for element (m, r) can be written as $\text{com}(m, r) = [m]G + [r]U = [m]G + [6r]G = [m + 6r]G$. Therefore, to fool Dmytro, it suffices to take such (m_2, r_2) that $m_2 + 6r_2 = m_1 + 6r_1$. Substituting numbers we get $15 + 6r_2 = 3 + 6 \cdot 7 = 45$. Therefore, $6r_2 = 30$ so finally $r_2 = 5$.

Exercise 3. Answer: D. It suffices to manually check every option. For option (D) we indeed have $H(H(17, H(3, 4)), 13) = 37$.

Exercise 4. Answer: C. The quotient polynomial $q(x)$ must satisfy $q(x) \cdot (x - 2) = p(x)$. It is easy to check that for $q(x) = x^2 - 8x + 15$ we indeed have $(x^2 - 8x + 15)(x - 2) = x^3 - 10x^2 + 31x - 30$.

Security Analysis

Exercise 1. Answer: B. Adversary $\mathcal{A}$ can safely send two different messages m_0 and m_1 to the challenger with $\text{lsb}(m_b) = b$ for $b \in \{0, 1\}$. Then, having received c , $\mathcal{A}$ can output $b' \leftarrow \mathcal{O}_{\text{lsb}}(c)$ where $\mathcal{O}_{\text{lsb}}(c)$ is an oracle that returns the least significant bit of m . Thus, $\Pr[b = \hat{b}] = 1$ and thus $\text{SSAdv}[\mathcal{A}] = 1/2$.

Exercise 2. Answer: D. The probability should be subtracted from the probability of random guess. The random guess corresponds to the probability $1/|\mathcal{M}|$, thus the answer.

Exercise 3. Answer: D. Take $f = g$. Then, $f/g = 1 \neq \text{negl}(\lambda)$.

Introduction to Zero-Knowledge Proofs

Exercise 1. Answer: A.

Exercise 2. Answer: B. The probability of accepting the false statement after n rounds is $(1/8)^n = 2^{-3n}$. To guarantee 120 bits of security, we need $3n = 120$ or $n = 40$ rounds.

Exercise 3. Answer: C. To check a great explanation, see the answer from [this forum](#).

Exercise 4. Answer: C.

Exercise 5. Answer: B.

Exercise 6. Answer: D.

Sigma Protocols

Exercise 1. The protocol is sound and complete, but it is not zero-knowledge.

Exercise 2. The protocol is not complete. Yet, it is sound and zero-knowledge.

Exercise 3. While the protocol is sound and complete, it is not zero-knowledge: based on (a, r) , the verifier can find α as $a - r$.

Exercise 4. The protocol is complete and zero-knowledge, but it is not sound. Indeed, the prover does not prove that r is chosen randomly, so he can send $(g^z/h, z)$ to the verifier, which would make the verifier accept the statement, yet the prover might not know the discrete logarithm of h .

Exercise 5. This is a Schnorr's protocol. It is sound, complete, and zero-knowledge.

Exercise 6. Answer: B. Note that $w = g^{\alpha\beta} = (g^\alpha)^\beta = u^\beta$.

Exercise 7. Answer: C. $v = g^\beta$ and $w = u^\beta$.

Exercise 8. Answer: C. $v_r = g^{\beta r}$ and $w_r = u^{\beta r}$.

Exercise 9. Answer: A. We need to hash both the initial statement (u, v, w) and the commitment (v_r, w_r) , so the hash function should input five group elements.

Exercise 10. Answer: A. $c \leftarrow H((u, v, w), (v_r, w_r))$.